Week 43

Table of contents

1 Commas and dots. 1
2 Concentration units 2
3 Calculating concentration from mass and volume 2
4 Mass calculated from concentration and volume 3
5 Molecular weight 3
6 Dilutions 3
7 Equilibrium reactions 3
8 Reaction data analysis 4
9 Buffer equation 4
10 The Henderson-Hasselbalch equation visualisering 5

```
try:
    import fysisk_biokemi
    print("Already installed")
except ImportError:
    %pip install -q "fysisk_biokemi[pyodide] @ https://au-mbg.github.io/fysisk-biokemi/
wheels/fysisk_biokemi-0.2.2-py3-none-any.whl"
```

```
import fysisk_biokemi.widgets as fsb_widgets
```

1 Commas and dots.

In Danish, commas are used as decimal separators, so 3.14 is written as 3,14. In English, dots are used as decimal separators, so the number is written as 3.14.

Commas can also have different meanings in different programs and file formats. For example:

- **Python**: Commas separate elements. Writing 3,14 in Python creates a tuple (a type of container) with two elements: 3 and 14.
- **CSV files**: In CSV files commas are used to separate values into columns. Therefore, a CSV file containing 3,14 on a single line may be interpreted as two columns: one containing 3 and the other containing 14.

This is important to keep in mind when working with data, as a decimal comma can be mistaken for a separator between values.

Try writing the following in the code cell below:

- `3,14 + 6`
- `3.14 + 6`

```
# Try your calculations here!!
...
```

What is the output of the codecell (what shows up) when you write '3,14 +6'? When you write '3.14+6'?

We will use "." (a dot) as the decimal separator from now on.

---

2 Concentration units

The widget below allows you to convert between units with different prefixes. The values of these prefixes are listed in the table below.

Unit	Name	Factor
M		1
mM	Milli	10^{-3}
µM	Micro	10^{-6}
nM	Nano	10^{-9}
pM	Pico	10^{-12}
fM	Femto	10^{-15}

Try out the widget. How many picomolar is 3.521 M?

```
fsb_widgets.concentration_unit()
```

3 Calculating concentration from mass and volume

When preparing experiments, you often need to calculate the concentration of a reagent based on the mass used and the final volume of the solution.

To determine the molar concentration, you will have to calculate the following:

$$c = \frac{m}{M_W \cdot V} \overset{\text{Enheder}}{\rightarrow} \frac{\text{g}}{\frac{\text{g}}{\text{mol}} \cdot \text{L}} = \frac{\text{g} \cdot \text{mol}}{\text{g} \cdot L} = \frac{\text{mol}}{\text{L}}$$

m is the mass in **gram**, M_W is the molecular weight in $\frac{g}{mol}$ and V er volume in liter It is important to use the correct units, as using inconsistent units can lead to incorrect results.

The widget below calculates the molar concentration using the mass of the reagent, the final volume of the solution, and the molecular weight. You can change the values and units of the mass, volume, and molecular weight to explore how these affect the calculated concentration. The calculation display shows how the result changes when you adjust the inputs.

Try the widget. How does the concentration change, when you change the unit of the mass, volume or molecular weight?

```
fsb_widgets.concentration_mass_volume()
```

4 Mass calculated from concentration and volume

When preparing a solution, you may know the desired molar concentration and the final volume but need to determine how much reagent to weigh out.

The widget below calculates the mass required to prepare a solution with a given concentration and final volume. You can adjust the concentration, volume, and molecular weight to see how each affects the required mass.

```
fsb_widgets.mass_concentration_volume()
```

5 Molecular weight

The molecular weight is the sum of the atomic weights of all the atoms in a molecule.

$$M_W = \sum_i^Z n_i m_Z$$

Here,n_i is the number of atoms of the element i and m_Z is the molar mass of that element. The widget below calculates the molecular weight of compounds made up of elements with atomic numbers 1–83 in the periodic table.

Enter the molecular formula of a compound and use the widget to calculate the molecular weight.

```
fsb_widgets.molecular_weight()
```

6 Dilutions

When preparing a dilution, you need to know the desired final concentration and volume, as well as how much stock solution and solvent to use.

The widget below calculates how much stock solution and solvent you need to prepare a solution at the desired concentration and final volume.

Try the widget. If you, for example, want 500 mL of a dilution that has a final concentration of 5 µM and your stock solution is 5 M. How much stock and solvent do you need to add?

```
fsb_widgets.dilution_helper()
```

7 Equilibrium reactions

The widget below can calculate the equilibrium constant of a reaction, for example:

$$aA + bB... = xX + yY...$$

Where the equilibrium constant is defined by:

$$K_{eq} = \frac{[X]^x \times [Y]^y \times ...}{[A]^a \times [B]^b \times ...}$$

In this example, it is (also) important that concentrations are expressed in mol/L to ensure that the equilibrium constant is calculated correctly. Fortunately, the widget can convert the units for you.

To use the widget, enter the reaction equation in the following form:

```
a A + b B = c C
```

Try out the widget. If the concentration of A is 5 mM, the concentration of B is 1000 µM and the concentration of C is is 0.2 M, what is the Equility constant?

```
fsb_widgets.reaction_equation()
```

8 Reaction data analysis

This widget can read a 'CSV'-file(**c**comma **s**seperated **v**calues -file). in a CSV file data are organised into columns, with commas seperating the values. An example of a CSV file is shown below.

```
{csv}
time (s),A (mM),B (µM),X (nM),Y (µM)
0,1.00,500.0,100.0,50.0
10,0.95,480.0,150.0,70.0
20,0.92,460.0,200.0,90.0
30,0.90,445.0,240.0,110.0
40,0.88,430.0,270.0,130.0
50,0.87,420.0,300.0,150.0
60,0.86,410.0,330.0,170.0
70,0.85,400.0,360.0,190.0
80,0.84,390.0,390.0,210.0
90,0.83,380.0,420.0,230.0
100,0.82,370.0,450.0,250.0
```

In this example, the CSV file contains 5 columns, each representing a different variable.

To use the widget, follow these four steps: 1. Click 'upload CSV' and select your file. 2. Make sure the reaction equation matches the reaction you want to analyse. 3. Make sure each component of the reaction equation is assigned to the correct column in your dataset, and ensure that the units are correct. 4. Click "Beregn og plot" to calculate the equilibrium constant.

```
fsb_widgets.reaction_data_analysis()
```

9 Buffer equation

The Henderson-Hasselbalch equation ('buffer equation') is defined by

$$pH = pK_a + \log_{10}\left(\frac{[\text{base}]}{[\text{acid}]}\right)$$

This widget uses the Henderson–Hasselbalch equation to calculate the ratio of conjugate base to acid, based on the pH and pKa and the total volume.

Try the widget. If the pH increases or decreases how does that affect the ratio of base to acid? If the pka increases/ decreases? If the total concentration changes?

```
fsb_widgets.buffer_equation()
```

10 The Henderson-Hasselbalch equation visualisering

This widget illustrates the The Henderson-Hasselbalch equation (the buffer equation). The plot shows the concentrations of the acid and its conjugate base as a function of pH. Change pKa, total concentration and pH to see howthese parameters affect the distribution of the acid and its conjugate base.

```
fsb_widgets.buffer_visualization()
```

```
HBox(children=(VBox(children=(FloatSlider(value=7.0, description='pKa:', max=14.0,
style=SliderStyle(descripti…
```